

# SAJAL REGMI

Tempe, AZ 85281



## EDUCATION

### Arizona State University

Bachelors of Science in Engineering (Biomedical Engineering)

August 2021 - May 2025

Tempe, Arizona

## SKILLS

### Languages

### Frameworks & Platforms

### Databases & Storage

### Software Architecture

TypeScript, Python, C++, C#, Objective-C++, JavaScript, SQL  
Node.js, React Native, Unity, PencilKit, Fabric, JSI, PyTorch  
PostgreSQL, MySQL, Redis, MongoDB, Neo4j  
Microservices, CQRS, event sourcing, message brokers, caching, REST APIs,  
gRPC/RPC, API security, RAG, GraphRAG, knowledge graphs, OT/CRDT,  
real-time collaboration  
AWS, ECS, Terraform, CI/CD, blue-green deployment, zero-trust security,  
infrastructure automation

### Cloud & Infrastructure

## OPEN SOURCE CONTRIBUTIONS

**GPT Semantic Cache** Package for caching LLM responses using semantic embeddings and similarity matching to reduce latency and inference cost.

**RN Markdown Editor** React Native package for editing and rendering Markdown content with equation and LaTeX support.

**Pencil Kit For React Native** Native bridge between iOS PencilKit and React Native with stroke-level control for real-time collaborative drawing.

**Transformer from Scratch** Ground-up PyTorch implementation explaining the transformer architecture that powers modern LLM systems.

## PREPRINT

GPT Semantic Cache: Reducing LLM Costs and Latency via Semantic Embedding Caching. <https://arxiv.org/pdf/2411.05276>

## EXPERIENCE

### Lead Engineer & CEO

Arkios.ai (arkios.ai) / Karela Technologies Inc.

July 2025 - Present

Mesa, Arizona, United States

- Leading engineering for Arkios, an enterprise AI platform for secure internal application building, workflow automation, and deployment of AI agents across business operations.
- Designed and built an in-platform application builder that allows enterprises to create internal applications on top of isolated, policy-controlled environments without exposing core infrastructure.
- Architected sandboxed client-side and server-side execution runtimes with tenant isolation, secure execution boundaries, and controlled access to tools, memory, and external systems.
- Built native document engine and call center modules to unify document workflows, knowledge access, human-in-the-loop operations, and execution flows inside a single platform.
- Developed a memory engine for enterprise-scale ingestion, normalization, indexing, and retrieval of organizational context across files, systems, and conversations within one platform.
- Designed a tool-calling protocol and foundry layer so non-technical teams can configure tools and workflows with limited API knowledge while preserving governance, auditability, and security controls.
- Led platform infrastructure and security architecture, including isolated tenant environments, provisioning automation, secrets management, distributed RPC between services and tenant nodes, and zero-trust deployment controls.

### Engineer

Arkios Code (code.arkios.ai) / Karela Technologies Inc.

January 2026 - Present

Mesa, Arizona, United States

- Building Arkios Code, a desktop firmware IDE for hardware engineers with integrated project workflows, flashing, serial console, crash analysis, RTOS inspection, and device debugging.
- Developed low-level, machine-independent tooling to unify debug and programming workflows across three MCU architectures, reducing per-target integration complexity.
- Designing custom debug probe hardware and host software to capture high-frequency waveforms, power measurements, and multimeter-style telemetry for board bring-up, validation, and failure analysis.
- Built C++ instrumentation and orchestration layers across probes, programmers, embedded toolchains, and lab equipment to create a common harness used by hardware engineers.
- Working on native desktop systems, debugger backends, and IDE integrations for symbol inspection, memory and register views, trace pipelines, and automated firmware development workflows.

## EXPERIENCE

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### Lead Software Engineer

January 2025 - July 2025

Lecturely AI

*Remote*

- Architected multi-agent LLM workflows with knowledge-graph-enhanced RAG, improving response relevance and context quality across the product experience.
- Built a React Native to native iOS bridge using Objective-C++, Fabric synchronization, and JSI bindings to enable low-latency PencilKit-based note and doodle features.
- Implemented real-time collaborative note-taking with an OT/CRDT hybrid model to maintain consistent multi-user state under concurrent edits.
- Automated testing and deployment on Amazon ECS using CI/CD, health checks, blue-green rollout strategy, and rollback automation.

### Entrepreneur in Residence

May 2024 - July 2024

Creatiphe LLC

*Remote*

- Built and validated MVPs through rapid experiments, improving activation and conversion by 30% using structured iteration and user testing.

### Co-Founder & Principal Engineer

May 2023 - May 2024

Ticketsewa Pvt Ltd

*Remote, Nepal*

- Architected and led a distributed ticketing platform using CQRS, event sourcing, and microservices to support high-concurrency purchase flows.
- Designed payment orchestration across airlines, events, and transport providers using idempotency and compensation patterns for resilient recovery.
- Built a real-time seat allocation engine with Redis locks and optimistic concurrency controls to prevent overselling under demand spikes.

### Product Development Intern

August 2020 - July 2021

Karyathalo IT and Web Solutions Pvt Ltd

*Kathmandu, Nepal*

- Delivered an e-governance platform for Kathmandu Metropolitan City using WordPress multisite and custom plugin architecture.
- Optimized MySQL indexing and Redis caching, reducing page load time from 4.2s to 1.1s under peak usage.

### Game Developer & Product Lead

February 2019 - February 2020

Karela Games

*Remote*

- Built and shipped Unity games in C#, improving acquisition through analytics-driven iteration and product experimentation.